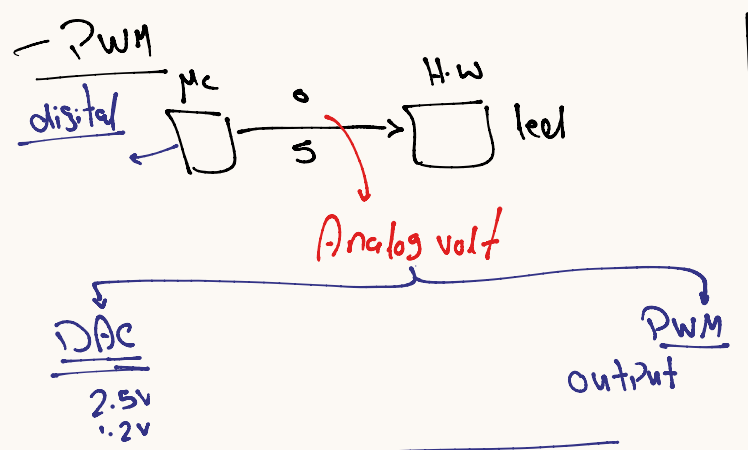
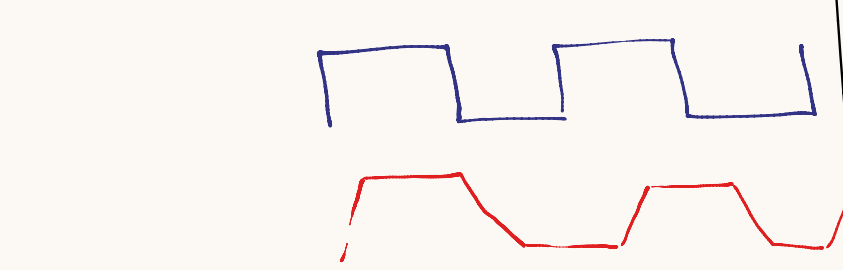


- Some Concept
- 1) Periodic time :: the time of 1 cycle
 - 2) Frequency :: No of Pulses in 1 sec

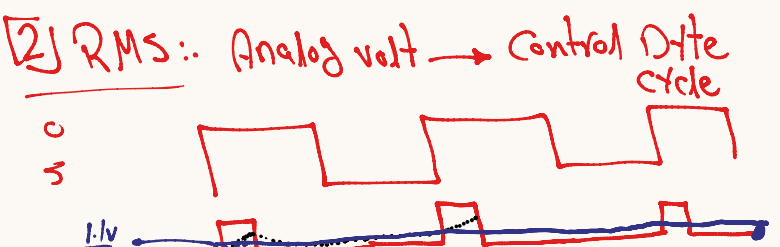
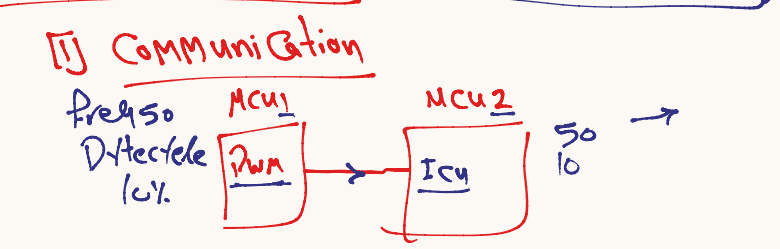
$$= \frac{1}{\text{Periodic}}$$
 - 3) on time :: time Pulse is High
 - 4) off time :: time Pulse is low
 - 5) Dte cycle :: Ratio between on time & Periodic
 - 6) Amplitude :: Max volt when signal is High

7) Root Mean Squared (RMS)

$$RMS = A \sqrt{\text{Duty cycle}}$$



PWM in Embedded



H.W generate PWM From Mc

by sw

→ ① Using delay.

```

while(1)
{
    Motor on ();
    delay_ms(300);
    Motor off ();
    delay_ms(700);
}

```

Calculation:

$$P_{\text{freq}} = \frac{1}{\text{Periodic}} = 1 \text{ kHz}$$

$$\text{Duty cycle} = \frac{300}{1000} \times 100 = 30\%$$

→ ② Using (Normal / CTC)

Normal mode → 2 Pins

Compare Reg 10 μsec

ISR (OVF)

```

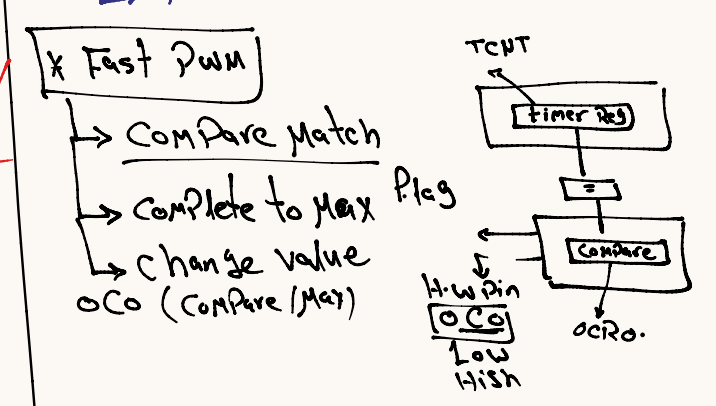
static unsigned char Counter = 0;
Counter++;
if (Counter == 30)
{
    set pin low;
}
else if (Counter == 100)
{
    Counter = 0;
}

```

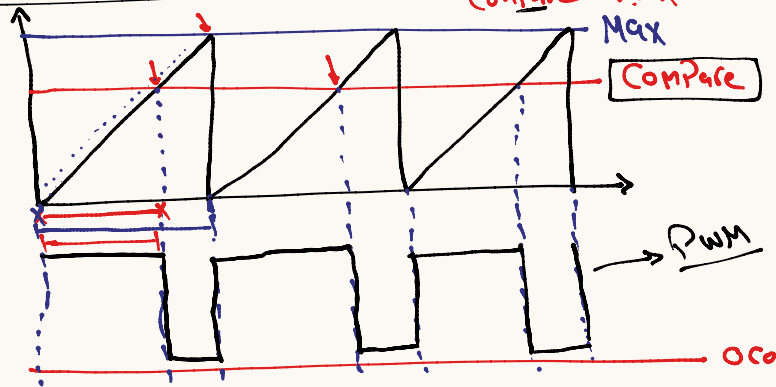
ISR (Compare)

* Generate PWM From Mc by H.W

- Fast PWM (Normal PWM)
- PWM Phase correct



* Fast PWM First Assume [clear | set] ^{Non Inverting}
Compare Max



* $\text{on time} = \text{compare time} = \text{clk time} \times \text{compare value} \text{ ①}$

* $\text{periodic time} = \text{overflow time} = \text{clk time} \times 2^n$

* $\text{Duty cycle} = \frac{\text{on time}}{\text{periodic time}} = \frac{\text{clk time} \times \text{compare value}}{\text{clk time} \times 2^n}$

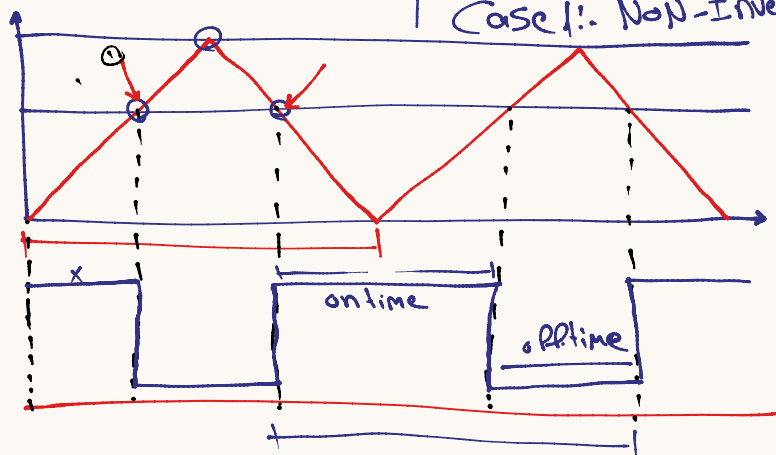
$\text{Duty cycle} = \frac{\text{Compare value}}{2^n}$

$\text{Compare value} = 2^n \times \text{Duty cycle}$

[Non-Inverting]

* Phase Correct PWM

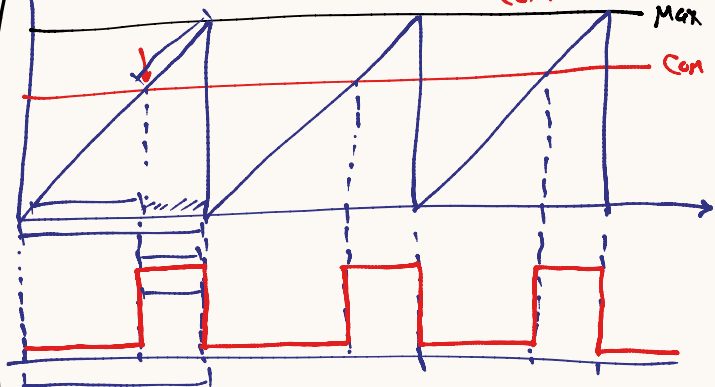
* low freq / long time → changes → down count flag
Case 1: Non-Inverting



* $\text{Duty cycle} = \frac{\text{compare value}}{2^n}$

$\text{compare value} = 2^n (\text{Duty cycle}) \rightarrow \text{Non-Inverting}$

* Fast PWM Second Assume [set | clear] ^{Compare Max}



$\text{on time} = \text{overflow time} - \text{compare time}$

$\text{Periodic time} = \text{overflow time}$

$\text{Duty cycle} = \frac{\text{on time}}{\text{Periodic}}$

$= \frac{\text{overflow time} - \text{compare time}}{\text{overflow}}$

$= \frac{\text{overflow}}{\text{overflow}} - \frac{\text{compare time}}{\text{overflow}}$

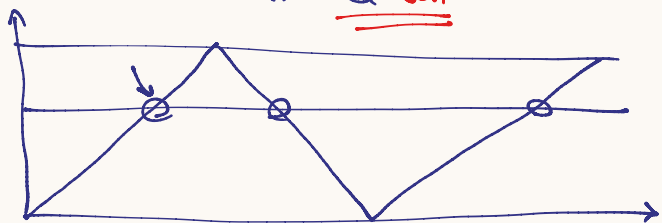
$= 1 - \frac{\text{clk time} \times \text{compare value}}{\text{clk time} \times 2^n}$

$\text{Duty cycle} = 1 - \frac{\text{compare value}}{2^n}$

$\text{compare value} = 2^n \times (1 - \text{Duty cycle})$

[Inverting Mode]

Case 2 [set | clear] ^{Inverting}
up down



$\text{compare value} = 2^n (1 - \text{Duty cycle})$